

Module 4 Summary – Jacob Booth 221530207

Response to feedback

No feedback yet from module 3, but I've tightened my new workflow for timeliness, with the additional aim of addressing concepts more to-the-point and emphasising underlying links. I've focused on adapting my writing style to align with standards for mathematical report writing, just hoping as always that I haven't written too summarily or briefly for anything that needs more focus. Other than that, I really enjoyed this module.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module:

- Calculate determinants using the cofactor expansion
- Use properties of determinants (Theorem 3.1.2) to calculate determinants
- Calculate matrix inverses using $\text{adj } A$ and $\det(A)$
- Identify the inverse of an orthogonal matrix
- Perform LU-factorisation on a matrix and use this to solve systems of linear equations

Key Definitions and Theorems

The Cofactor Expansion

Textbook Ref	Theorem/Definition statement	Learning Notes
Determinant Calculation for a 3×3 matrix	For a 3×3 matrix, A – the <u>determinant</u> is given by $\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $= aei + bfg + cdh - ceg - afh - bdi$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$ $= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$	Draws on the products of diagonals Use either the shortcut formula with Sarrus' rule or expansion by minors
Definition 3.1 Cofactors of a matrix	Assume that determinants of $(n - 1) \times (n - 1)$ matrices have been defined. Given the $n \times n$ matrix A, let A_{ij} denote the $(n - 1) \times (n - 1)$ matrix obtained from A by deleting row i and column j. Then the (i, j)-<u>cofactor</u> $c_{ij}(A)$ is the scalar defined by	Dissecting sub-matrices from larger ones, carrying the idea of composition Convenient method, just requires reference to the formula

	$c_{ij}(A) = (-1)^{i+j} \det(A_{ij})$ Here $(-1)^{i+j}$ is called the <u>sign</u> of the (i, j)-position.	
Definition 3.2 Cofactor expansion of a matrix	Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. If $A = [a_{ij}]$ is $n \times n$ define, $\det A = a_{11}c_{11}(A) + a_{12}c_{12}(A) + \dots + a_{1n}c_{1n}(A)$ This is called the <u>cofactor expansion</u> of $\det A$ along row 1.	Defining the determinant from smaller pieces, recursively defining smaller matrix dets to define the det of A
Theorem 3.1.2 Determinant properties	Let A denote an $n \times n$ matrix. 1. If A has a row or column of zeros, $\det A = 0$. 2. If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is $-\det A$. 3. If a row (or column) of A is multiplied by a constant u, the determinant of the resulting matrix is $u(\det A)$. 4. If two distinct rows (or columns) of A are identical, $\det A = 0$. 5. If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is $\det A$.	Unique behaviour of det, proportional scaling but adding one row to another doesn't change det Reduces the logic down to 5 constraints on the determinant, helpful for quick interpretation

Determinants and matrix inverses

Textbook Ref	Theorem/Definition statement	Notes
Theorem 3.2.1 Product theorem	If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B.$	Matrix product det or det of matrix product, both are the same
Theorem 3.2.2 Det relationship to inverse	An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When it is invertible, we have, $\det(A^{-1}) = \frac{1}{\det A}$	Any matrix with a non-zero det is invertible
Theorem 3.2.3 Det of transpose	If A is any square matrix, $\det(A^T) = \det A$	Det persists across transposition
Theorem 3.2.4 Adjugate formula	If A is any square matrix, then $A(\text{adj } A) = (\det A) I = (\text{adj } A) A$ In particular, if $\det A \neq 0$, the inverse of A is given by $A^{-1} = \frac{1}{\det A} \text{adj } A$	Multiplying A with $\text{adj}(A)$ gives $(\det(A))I$ Leading into the inverse formula

LU Factorisation

Textbook Ref	Theorem/Definition statement	Notes
Theorem 2.7.1 LU Theorem	If A can be lower reduced to a row-echelon matrix U, then $A=LU$ where L is lower triangular and invertible and U is upper triangular and row-echelon.	Without row swaps, if matrix A can be reduced then it factors into L and U composition
LU Algorithm	Let A be an $m \times n$ matrix of rank r, and suppose that A can be lower reduced to a row-echelon matrix U . Then $A = LU$ where the lower triangular, invertible matrix L is constructed as follows: 1. If $A=0$, take $L=I_m$ and $U=0$. 2. If $A \neq 0$, write $A_1=A$ and let $\mathbf{c}_1$ be the leading column of A_1. Use $\mathbf{c}_1$ to create the first leading 1 and create zeros below it (using lower reduction). When this is completed, let A_2 denote the matrix consisting of rows 2 to m of the matrix just created. 3. If $A_2 \neq 0$, let $\mathbf{c}_2$ be the leading column of A_2 and repeat Step 2 on A_2 to create A_3. 4. Continue in this way until U is reached, where all rows below the last leading 1 consist of zeros. This will happen after r steps. 5. Create L by placing $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_r$ at the bottom of the first r columns of I_m.	For each multiplier in elimination, build L from each m, recording scalar pivots in L and reduced A in U

Summarising your understanding:

Module 4 ties together theory from Gaussian Elimination, matrix multiplication, and LU Theorem, outlining the behaviour of determinants in the scope of cofactors and compositions. The module underpins cofactor expansion as a foundation for determinant calculation, and compares the preservation/scaling properties of each elementary operation. The matrix adjugate and transposition are reintroduced to highlight the behaviour of the determinant across different forms of a matrix.

4.1 The Cofactor Expansion

Determinant Calculation for a 3x3 matrix

Based on 3x3 square matrices, the determinant can be calculated via the explicit formula $\det(A) = aei + bfg + cdh - ceg - afh - bdi$, or by cofactor expansion, breaking A down into smaller parts and building the determinant from sub-matrix determinant results.

Definition 3.1 Cofactors of a matrix

Referred to as a signed determinant of position (i, j) , finding the cofactor of a row i , column j entry involves removing the row and column before taking the determinant from the smaller remaining matrix. Multiplying the result by $(-1)^{i+j}$ adjusts the +/- sign for the final result.

Definition 3.2 Cofactor expansion of a matrix

Building from cofactors as signed determinants in **Definition 3.1**, cofactor expansion is used for calculating the determinant of a matrix via its cofactors, as opposed to the formula stated in determinant calculation for 3×3 matrices. This involves multiplying each row/column entry by its cofactor (D3.1), giving the determinant.

Theorem 3.1.2 Properties of determinants

Drawing on structural behaviour determined by **Definitions 3.1, 3.2**, this theorem outlines the defining characteristics of determinants, dictating whether the determinant will be 0 or nonzero, and whether it will be a positive or negative number. Scalar row operations can scale the determinant while adding a multiple of one row to another doesn't change it at all. Looking specifically at condition 4, $\det(A)$ becomes 0 due to cofactor expansion along that row, cancelling out with its opposite corresponding $(-1)^{i+j}$ contribution (D3.1) when cofactors are summed (D3.2). This indicates a recursive relationship between D3.1, 3.2, where each determinant requires cofactor expansion until the base case (smallest component) is reached.

N4.2.2

When $\det(A) = 9$, find $\det(B)$

Swap 1 $\leftrightarrow$ 4, 2, 3 unchanged

$$\therefore -\det(A) \Rightarrow \det(B) = -\det(A) = -9$$

Using Theorem 3.1.2

condition 2, $-\det(A)$

N4.2.3

$A = \begin{pmatrix} -2 & 0 & -3 & 2 \\ 4 & 2 & 5 & -7 \\ 0 & -2 & -1 & -3 \\ 6 & 0 & 4 & 5 \\ 0 & 0 & 2 & -1 \\ 2 & 6 & 2 & 14 \\ 0 & 6 & 1 & -12 \end{pmatrix}$

Step 1: $R_1 \leftrightarrow R_7$, $R_2 \leftrightarrow R_6$, $R_3 \leftrightarrow R_5$

Step 2: $R_1 \leftarrow R_1 + R_2$, $R_3 \leftarrow R_3 + R_4$, $R_5 \leftarrow R_5 + R_6$

Step 3: $R_1 \leftarrow R_1 - R_2$, $R_3 \leftarrow R_3 - R_4$, $R_5 \leftarrow R_5 - R_6$

$$\therefore \det(A) = (-2)(-2)(-3)(-4) = 32$$

Property 5, $\det$ unaltered

A worked example of applying rule 2 of Theorem 3.1.2 to simplify the calculation of a larger determinant, followed by an example applying rule 5 to persist the determinant through elimination (no scaling) for simplicity.

4.2 Determinants and Matrix Inverses

Theorem 3.2.1 Product Theorem

The base theorem for composition with determinants (Sub-module 4.1), multiplying two square matrices and taking their determinant will be the same as taking the determinant of each and multiplying them, both are equal. Determinants are subject to predictable behaviour through multiplication, and are convenient for linear analysis. Product Theorem underpins behaviour outlined by **Theorem 3.2.2 → 3.2.4** when exposing determinants of square matrices, their inverses, transpositions, and adjugates, to determinant calculations.

Theorem 3.2.2 Determinant relationship to inverse

The logic applied to **Product Theorem 3.2.1** infers the inverse for a matrix A will always exist given that the determinant isn't zero. A zero determinant "flattens" the matrix, collapsing it in a way that can't be reversed to retrieve the original information, it's all lost during cofactor expansion as it reduces to 0. This is consistent with conditions outlined by **Theorem 3.1.2**, for example:

2) rows or columns swapped $\rightarrow \det(A) = -\det(A)$,

4) matching cols or rows $\rightarrow \det(A) = 0$.

A non-zero determinant indicates full rank and the existence of an inverse.

Theorem 3.2.3 Determinant of transpose

Flipping the matrix over its diagonal won't alter the determinant. Persistence over transposition demonstrates that row-based and column-based operations are symmetric in regards to determinants.

The transpose is equivalent to the inverse for an orthogonal matrix A . The determinant of a transposed orthogonal matrix is therefore the same as the determinant of the inverse orthogonal matrix, both of which are equal to the determinant of the original system.

Theorem 3.2.4 Adjugate Formula

Applying **Definitions 3.1, 3.2** for building the adjugate from determinants and cofactors, the adjugate formula forms $\text{adj}(A)$ by transposing the cofactor matrix of A . This depends on cofactor expansion being consistent through transposition of A , as outlined by **Theorem 3.2.3**.

Cramer's Rule is a method for solving a linear system $Ax = b$, given $\det(A)$ is a non-zero value (T3.2.2) and the coefficient matrix is square. For each x_i in the system, replace the column i of A with the constant vector b , giving an intermediate matrix A_i . This gives the solution $x_i = \det(A_i) / \det(A)$. Cramer's rule draws on the behaviour established by the adjugate formula (T 3.2.4). The solution for x is $\text{inverse-}A$ multiplied by constant vector b , which is the same as $(1/\det(A)) * \text{adj}(A) * b$. Each component draws on cofactor expansion with a modified matrix to establish an explicit formula for each variable.

4.3 LU Factorisation

Theorem 2.7.1 LU Theorem

Drawing on the same underlying logic for cofactor expansion (D3.2) without applying it explicitly, LU Theorem implies that given a matrix A can be reduced to row-echelon form without using any row swaps, A can be expressed as a matrix product composed of two smaller matrices, L and U . L is an invertible $m \times m$ matrix that holds the lower-triangular matrix of only non-zero values below the diagonal, corresponding to row op scalars. U is an upper-triangular $m \times n$ matrix in row-echelon form, denoting the reduced A . Given the decomposition of A , into row op scalars L , and the row-echelon matrix U , the composition $A = LU$ holds true. Consistent with **Theorem 3.1.2** property 5, avoiding any row/col swaps doesn't alter the determinant.

LU Algorithm

Reducing A to row-echelon form through Gaussian Elimination, record the scalars used for elimination below the pivot into L , using the corresponding identity matrix as the foundation. The resulting U matrix is the same as A after elimination. This establishes the step-by-step method for factorising a matrix A into LU . LU Algorithm relies on the determinant behaviours outlined by **Theorems 3.1.2, 3.2.1**, demonstrating:

- Determinant consistency across row addition
- The product using determinants of matrices is consistent with the single determinant of a matrix product.

LU factorisation can be applied to full-rank square matrices, so long as they can be reduced to an upper-triangular without any row swaps. LU Factorisation decomposes a square matrix A into the product of a lower-triangular matrix L and upper-triangular matrix U , expressing A as a factor of LU . This simplifies the problem space for solving $Ax = b$ by reducing the matrix A into two smaller triangular systems, easier for solving. Additionally, passing in any constant vector b to solve for new solutions can be done with the same L and U matrices, speeding up solving for different constant term sets.

Module 4 encapsulates the recursive nature of determinant calculations, the relationships between determinant consistency and elementary row operations, and the behaviour of the determinant across different factorisations. The determinant provides a convenient metric for analysing a system's properties and characteristics, like scaling behaviour, rank, etc.. LU factorisation ties this all together into a single workflow, demonstrating the efficient nature of solving linear systems as a composition of simpler parts. Respecting the constraints defined in this module, larger systems can be broken down without altering their defining properties, making solving and analysis easier and more efficient going forward.

Reflecting on the content

I found this module more intuitive than it initially appeared. I received a hit of nostalgia performing cofactor expansion and 3×3 determinant calculation, methods I'd seen and used in past classes without establishing the link until I was solving the problems. This was helpful and encouraging. I can also relate to the necessity of these concepts as a problem-solver

from a computer science background. Breaking down large problems into meaningful, functional components forms the backbone of robust software engineering, in addition to the key module concepts being directly applicable to linear data analysis and software speedup. Overall, a very enjoyable unit that simplifies all the granular processes we've been working with weeks 1 – 4.

References

Summary, notes

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Verification

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